Infinite NPA depth at minimal semidefinite complexity

Anubhav Chaturvedi[1]

[1] *Division of Quantum Optics and Information, Institute of Theoretical Physics and Astrophysics, Faculty of Mathematics, Physics and Informatics, University of Gdańsk, 80-308 Gdańsk, Poland*

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The quantum region of the CHSH value and the sum of two marginal averages requires infinite Navascués–Pironio–Acín depth, yet admits an exact finite semidefinite lift. We construct this lift by showing that every supporting bound is attained with equal measurement overlaps: one angle determines the entire boundary. Finite NPA certificates must remain valid for two independently varying angles, where the obstruction to finite depth survives. The same region has the smallest semidefinite extension degree possible for a nonpolyhedral set, separating infinite hierarchy depth from minimal semidefinite complexity.

Alice and Bob each choose between two measurements with outcomes ±1. Write $a_x = \langle A_x\rangle$, $b_y = \langle B_y\rangle$ and $E_{xy} = \langle A_x B_y\rangle$, and retain

$$M = a_0 + b_0, \qquad S = E_{00} + E_{01} + E_{10} - E_{11}. \quad (1)$$

S is the CHSH value [1]. Let $\mathcal{W}$ be their full quantum region. The two marginal averages may differ; only their sum is retained. Mixing strategies makes $\mathcal{W}$ convex, and the two-qubit reduction below makes it compact.

The preceding Letter [2] proved that every finite NPA level [3, 4] overestimates some supporting lines $S + \alpha M$ near $\alpha = 1$. Here we construct an exact finite lift of this same region. Write $\mathcal{W}_L$ for its level-L NPA projection. Its semidefinite extension degree—the smallest possible largest PSD block order in a finite lift—is two:

$$\mathcal{W} \subsetneq \mathcal{W}_L \quad \text{for every finite } L, \qquad \mathrm{sxdeg}(\mathcal{W}) = 2. \quad (2)$$

The explicit construction below uses blocks of orders five and four. The optimal order two follows from the geometry of the region.

Equal overlaps determine every supporting line.— To find $\mathcal{W}$, fix a candidate inequality $hS + \alpha M \le \lambda$. For given measurements its validity in every state is equivalent to

$$\lambda I - H \succeq 0, \qquad H = h\,\mathrm{CHSH} + \alpha(A_0 + B_0).$$

After projective dilation, Jordan's lemma decomposes each pair of binary observables into blocks of dimension at most two. Choosing local axes within these blocks gives

$$\begin{aligned} A_0 = B_0 = Z, \\ A_1 = cZ + \sqrt{1-c^2}X, \quad B_1 = dZ + \sqrt{1-d^2}X. \end{aligned} \quad (3)$$

The cosines $c, d \in [-1, 1]$ are the local measurement overlaps. Deterministic strategies attain $\ell = 2|h| + 2|\alpha|$, so a valid bound must satisfy $\lambda \ge \ell$.

Above this bound, the determinant decides positivity of the slack. To see this, put $\Omega = 1 + c + d - cd \in [-2, 2]$ and expand $\det(rI + \lambda I - H) = r^4 + e_1 r^3 + e_2 r^2 + e_3 r + D$:

$$\begin{aligned} e_1 &= 4\lambda, \qquad e_2 = 6\lambda^2 - 4\alpha^2 - 8h^2, \\ e_3 &= 4\lambda^3 - 8\lambda\alpha^2 - 16\lambda h^2 - 8h\alpha^2\Omega. \end{aligned} \quad (4)$$

These three coefficients are nonnegative. For e_3, replacing its last term by $-16|h|\alpha^2$ gives an increasing cubic on $\lambda \ge \ell$, whose value at ℓ is $16|\alpha|(|\alpha| + 2|h|)^2$. If $D \ge 0$ too, the polynomial is positive for every $r > 0$; a negative eigenvalue of the slack would give such a root.

We can therefore compare measurement choices through a scalar. Set $x = (1 - c)/2$, $y = (1 - d)/2$, and write $D(x, y) = \det(\lambda I - H)$. Direct expansion gives

$$D(x, y) = F(xy) + 64h^2(4h^2 - \alpha^2)xy(x + y), \quad (5)$$

where F depends only on the product. For $|\alpha| \le |h|$, the coefficient of the sum is nonnegative. At fixed product, $x + y \ge 2\sqrt{xy}$, so

$$D(x, y) \ge D(\sqrt{xy}, \sqrt{xy}). \quad (6)$$

The equal-overlap point remains in $[0, 1]^2$. Positivity there implies positivity at the original pair. Thus equal overlaps are the hardest measurements on which to satisfy the supporting bound.

For $|\alpha| \ge |h|$, the support is simply ℓ. The endpoint tilted-CHSH inequality follows from the identity

$$\begin{aligned} 4I - (\mathrm{CHSH} + 2A_0) = (I - C - A_1 D_B)^2 \\ + \tfrac{1}{4}\sum_{y=0}^{1}[(I - A_0)(I + B_y)]^2, \end{aligned} \quad (7)$$

where $C = (B_0 + B_1)/2$ and $D_B = (B_0 - B_1)/2$. Party exchange and output flips give either sign of CHSH and either sign of the marginal. Average the Alice and Bob bounds, multiply by $|h|$, and add the remaining marginal slack of weight $|\alpha| - |h|$. This proves $H \le \ell I$, attained with equal-overlap deterministic measurements. Equal overlaps therefore suffice for every support direction.

One angle gives one finite lift.— At $c = d$, exchange symmetry separates the singlet, whose eigenvalue is $-2hc$. On the other three Bell states, set $c = (1 - t^2)/(1 + t^2)$ and multiply the slack by $(1 + t^2)^2$. This removes the square root and gives

$$K(t) = \begin{pmatrix} \lambda p_0 - 2hp_1 & -2\alpha p_0 & 0 \\ -2\alpha p_0 & \lambda p_0 - 2hp_2 & -8ht^3 \\ 0 & -8ht^3 & \lambda p_0 + 2hp_2 \end{pmatrix}, \quad (8)$$

with $p_0 = (1+t^2)^2$, $p_1 = 1-t^4$ and $p_2 = 1+4t^2-t^4$. At $t=0$ and in the leading coefficient, positivity forces $\lambda \geq 2h + 2|\alpha|$ and $\lambda \geq -2h + 2|\alpha|$. These imply $\lambda \geq \ell$ and control the singlet. Continuity includes $c = -1$, while $K(-t) = RK(t)R$, $R = \mathrm{diag}(1,1,-1)$. Consequently, for all real support coefficients,

$$hS + \alpha M \leq \lambda \text{ on } \mathcal{W} \quad \Longleftrightarrow \quad K(t) \succeq 0 \quad \forall t \in \mathbb{R}. \tag{9}$$

The infinitely many angle tests now have a finite certificate. A PSD real univariate matrix polynomial is a polynomial matrix sum of squares, including when singular [6]. Quartic diagonal entries force every factor to have degree at most two, since leading real squares cannot cancel. Thus the same finite feature list works for every supporting line. For $\xi \in \mathbb{R}^3$, average a Gram representation under $(t,\xi) \mapsto (-t, R\xi)$. Mixed parities cancel, leaving the two feature vectors

$$Z_+ = (\xi_1, \xi_2, t\xi_3, t^2\xi_1, t^2\xi_2)^T,$$
$$Z_- = (\xi_3, t\xi_1, t\xi_2, t^2\xi_3)^T.$$

Equation (9) is therefore equivalent to the polynomial identity

$$\xi^T K(t) \xi = Z_+^T G Z_+ + Z_-^T J Z_-, \qquad G, J \succeq 0. \tag{10}$$

To turn these certificates into a membership test, replace evaluation of their monomials by a linear functional. The entries of $Z_+ Z_+^T$ and $Z_- Z_-^T$ span a sixteen-dimensional polynomial space. Let $\mathcal{L}_z$ assign real values $z \in \mathbb{R}^{16}$ to a basis of that space, and set

$$U(z) = \mathcal{L}_z(Z_+ Z_+^T), \qquad V(z) = \mathcal{L}_z(Z_- Z_-^T).$$

Define the three linear coordinate maps by

$$\mathcal{L}_z(\xi^T K(t) \xi) = \lambda \ell(z) - \alpha m(z) - h s(z).$$

If $U(z), V(z) \succeq 0$ and $\ell(z) = 1$, applying $\mathcal{L}_z$ to Eq. (10) gives

$$\lambda - \alpha m(z) - h s(z) = \mathrm{tr}(UG) + \mathrm{tr}(VJ) \geq 0. \tag{11}$$

Thus $(m(z), s(z))$ satisfies every supporting inequality of $\mathcal{W}$.

Conversely, each real unit triplet state ξ at angle t supplies the functional $P \mapsto P(t,\xi)/(1+t^2)^2$. It gives

$$U = \frac{Z_+ Z_+^T}{(1+t^2)^2}, \qquad V = \frac{Z_- Z_-^T}{(1+t^2)^2}, \tag{12}$$

with normalization one and coordinates equal to the state's actual M, S. The scaled features have limits at $t = \infty$. These triplet images form a compact set. Their convex hull has every quantum support value: equal overlaps suffice, and each singlet point $(0, -2c)$ is a mixture of the triplet endpoint points $(0, \pm 2)$. Their convex hull is therefore $\mathcal{W}$. Mixing the corresponding functionals gives

$$\mathcal{W} = \{(m(z), s(z)) : \ell(z) = 1,\ U(z), V(z) \succeq 0\}. \tag{13}$$

Both inclusions follow directly, including the boundary. The rational matrices are listed in the complete proofs [5]. Moreover, the triplet image is connected: every point of its planar convex hull lies on a chord between two image points. Thus two real equal-overlap strategies suffice for every (M, S), and both membership matrices can simultaneously have rank at most two [5].

Infinite depth, minimal block order.— The equal-overlap reduction concerns support values. Finite NPA certificates must hold for independently varying measurement angles. Near $\alpha = 1$, the preceding proof [2] shows that a NPA error bounded relative to the quantum violation would give a polynomial sum of squares for

$$\mathfrak{m}(X,Y) + c, \qquad \mathfrak{m} = 1 + X^4Y^2 + X^2Y^4 - 3X^2Y^2, \quad c \geq 0.$$

This is impossible: constant values on both axes restrict each cubic square factor to $a + bXY + dX^2Y + eXY^2$, whose squared X^2Y^2 coefficient is nonnegative. Yet on equal angles,

$$\mathfrak{m}(X,X) = (X^2-1)^2(2X^2+1).$$

The obstruction needs the freedom discarded by the support reduction. The complete scaling and limiting argument is in the companion [5].

For $f(\alpha) = \max_{\mathcal{W}}(S + \alpha M)$, the quantum boundary satisfies [2, 7]

$$f(1-T) = 4 - 2T + \tfrac{4}{3}T^3 + O(T^4), \qquad T \downarrow 0.$$

A polygon would have affine support sufficiently near this direction, so $\mathcal{W}$ is nonpolyhedral. Its finite lift makes it semialgebraic. Scheiderer's theorem gives extension degree at most two for every closed convex semialgebraic set in the plane [8]; order-one lifts are polyhedral. Hence $\mathrm{sxdeg}(\mathcal{W}) = 2$.

The construction also gives thirty-two exact projections. For each CHSH sign pattern χ with $\prod_{ij} \chi_{ij} = -1$, and each selected input pair x, y, physical relabellings identify

$$\left(a_x + \chi_{xy} b_y,\ \sum_{ij} \chi_{ij} E_{ij} \right)$$

with the same region $\mathcal{W}$. Adding these lifts to any NPA level retains every quantum correlation and makes all supporting bounds on these planes exact, including the family responsible for finite-level nontermination.

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[5] See the accompanying Supplemental Material for the complete finite-lift proof, explicit membership matrices, all signed asymmetric support branches, physical relabellings and the full earlier NPA proof.

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